

Sarbajit Ghosh

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SUMMARY

Electronics and Communication Engineering student passionate about building impactful technology and solving real-world problems through research, innovation challenges, and hands-on projects. Curious, adaptable, and committed to continuous learning across disciplines.

WORK EXPERIENCE

Research Intern, I2CS Research Lab, IIIT Kottayam

May 2026 - July 2026

- Designed and implemented a retraining-free structured pruning pipeline for CNNs (ResNet-56, VGG-16) using L1-norm filter ranking and BatchNorm statistics recalibration, eliminating the fine-tuning stage entirely.
- Built the complete PyTorch experimental codebase from scratch and evaluated it across multiple sparsity levels (10–80%) and 3 random seeds per architecture for statistically robust results.
- Achieved 82.90% Top-1 accuracy at 50% sparsity on ResNet-56/CIFAR-10, improving naive pruning by 26.2 percentage points using only 1,024 unlabelled calibration images and under 0.5s per sparsity level.
- Outperformed existing retraining-free pruning approaches at moderate-to-high sparsity, validated through a multi-seed ablation study across two CNN architectures.

AI Intern, Work2Hire (Rozgar Career Tech Pvt. Ltd.)

May 2026 - July 2026

- Built the core pipeline and model integration for an AI-powered ATS resume generation system using local LLMs with zero external API dependency.
- Developed a multi-agent pipeline of 6 agents automating resume intake, JD gap analysis, summary and bullet rewriting, project enhancement, keyword injection, and QA validation.
- Built a 5-component ATS scoring model combining BM25 keyword matching and Sentence-BERT semantic similarity to score resumes against job descriptions in real time.
- Engineered a modular LLM backend supporting three interchangeable model sources and curated a domain-specific prompt library spanning 80+ job roles for tailored resume generation.

Winter Intern, India Space Lab

Feb 2026 - Mar 2026

- Completed structured technical training across advanced space-technology domains including drone technology, CanSat/CubeSat satellite systems, remote sensing and GIS, rocketry, and disaster management.
- Designed a Dual MCU + Dual Sensor + Dual Servo PCB and CAD project, applying embedded systems design and hardware layout concepts.

PROJECTS

AI-Driven Panic Detection & Emergency Response System

Tech Stack: SIM7600E, MAX30102, MPU6050, GSR, PyTorch, CNN-LSTM, Knowledge Graph, Geofencing, Embedded C

- Built a wearable prototype using SIM7600E for 4G LTE SOS dispatch and GNSS/GPS, integrated with MAX30102 PPG, GSR, and MPU6050 IMU sensors.
- Integrated the sensor pipeline with the AI/ML inference stack, contributing to a hybrid CNN-LSTM + Knowledge Graph + Geofencing framework achieving 96.8% accuracy, 96.2% recall, and 0.978 AUC.
- Co-authored the accompanying research paper detailing a six-layer architecture combining wearable sensing, deep learning, knowledge-graph reasoning, and geofenced risk mapping.

Smart Energy Forecasting & Anomaly Detection System

Tech Stack: ESP32, Current Sensor, Voltage Sensor, Python, Pandas, NumPy, Scikit-learn, Random Forest

- Built an end-to-end ML pipeline on smart meter data to forecast electricity consumption, detect abnormal usage, and quantify energy wastage.
- Engineered temporal features and trained regression and classification models, using class-weight balancing to achieve 96% anomaly-detection accuracy with perfect precision.
- Designed a rule-based corrective action engine that generates alerts and equipment-inspection recommendations based on detected wastage events.

SKILLS

Programming: C, C++, Python
AI/ML: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, PyTorch, CNN, RNN, LSTM, LLMs, NLP, Generative AI, Supervised Learning, Regression, Classification, Feature Engineering, Model Evaluation
Embedded & IoT: Arduino, ESP32, Raspberry Pi, PCB Design & Fabrication, Sensor Integration
Tools: Git, VS Code, Jupyter Notebook, Arduino IDE, EasyEDA, Fritzing, Xilinx, Cadence, TCAD

EDUCATION

2024 - 2028	Birla Institute of Technology, Mesra, Ranchi B.Tech, Electronics & Communication Engineering	(CGPA: 8.28/10)
Class XII	Manisha International School, Durgapur, West Bengal	(CBSE: 79.4%)
Class X	West End High School, Jhargram, West Bengal	(ICSE: 94%)

CERTIFICATIONS

- Introduction to Generative AI — Google Cloud (Coursera), July 2026
- Introduction to Deep Learning & Neural Networks with Keras—IBM (Coursera), June 2026
- Machine Learning with Python—IBM (Coursera), July 2026

PUBLICATIONS

A. Gaur, Y. Waghela, **S. Ghosh**, S. Shekhar, and P. Gupta. *AI-Driven Panic Detection and Emergency Response Framework Using Physiological Signal Analytics, Hybrid Deep Learning Models, Knowledge Graph Intelligence and Geofenced Risk Mapping*. Presented at the 6th International Conference on Data Science and Big Data Analytics (IDBA-2026), Symbiosis University of Applied Sciences.

ACCOMPLISHMENTS & VOLUNTEERING

- Co-authored and presented a research paper, *AI-Driven Panic Detection and Emergency Response Framework*, at the 6th International Conference on Data Science and Big Data Analytics (IDBA-2026), Symbiosis University of Applied Sciences.
- Won **1st Position** at IETE Students' Innovation Day for EDU-RESQ, an RFID-based smart resource allocation system.
- Secured **2nd Position** at BIT Mesra Internal Hackathon for innovative project development and implementation.
- Coordinator, AI Club — organized technical sessions on AI and emerging tech.
- Coordinator, National Service Scheme (NSS) — planned and executed community service and social outreach initiatives.